PRIMA
PARTNERSHIP FOR RESEARCH AND INNOVATION IN THE MEDITERRANEAN AREA

This project is part of the PRIMA programme supported by the European Union

CRS4 IDEAS BECOME LIFE

soils4med

SOIL health monitoring and information systems FOR sustainable soil management in the MEDiterranean region

MedSIS manual

USER Guide

Roberto Demontis, Eva Lorrai, Mais Atiyat



Authors: Roberto Demontis (demontis@crs4.it), Eva Lorrai (eva@crs4.it), Mais Atiyat (mais.atiyat@hotmail.com)

Delivery dates: **10 Settembre 2026**

Table of contents:

1. Overview **4**
2. First Access **4**
3. User interface **5**
The resources list panel 6
Dataset’s resource panel 8
Map’s resource panel 11
Documents’s resource panel 14
Geostory’s resource panel 14
Dashboard’s resource panel 14
4. How To Requests Data Access **15**

1. Overview

This document describes the general use of the SIS platform for a generic user.

The MED-SIS is essentially a catalogue of soil data resources based on Geonode (https://geonode.org/ see the documentation at https://docs.geonode.org/en/latest/).

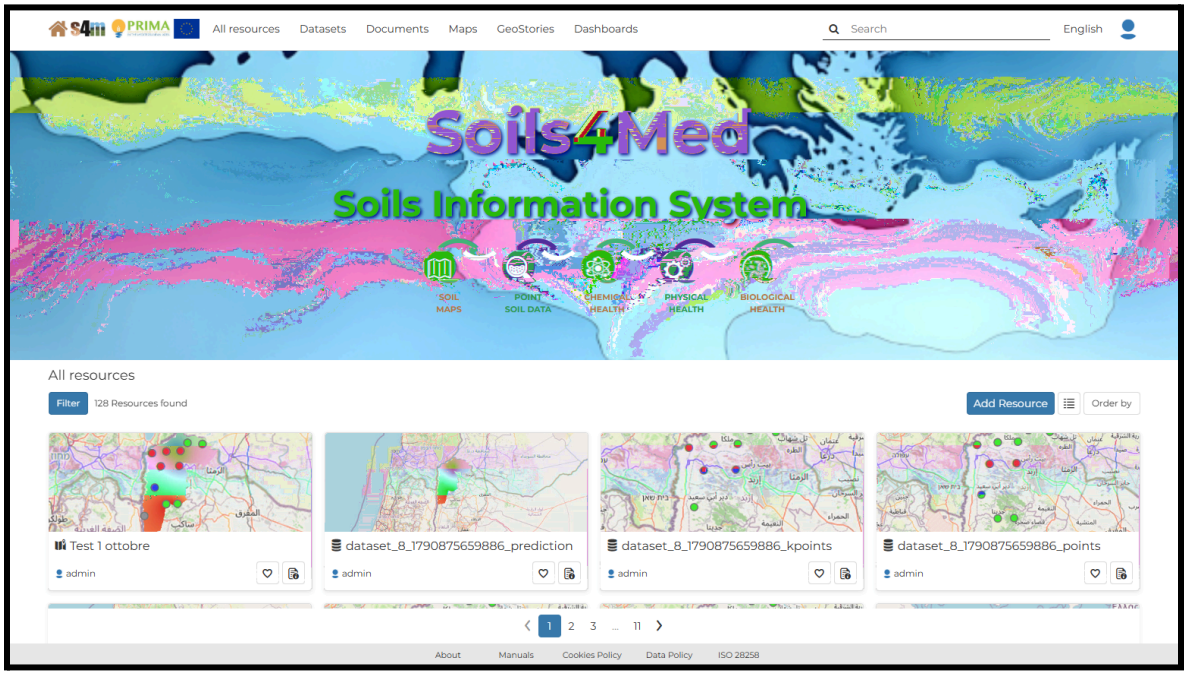


Figure 1: The home page



Figure 2. Bottom bar

The bottom bar of all pages shows some link:

- **About**: to display site description
- **Manuals**: to display help page
- **Cookies** Policy: to display cookies policy
- **Data Policy:** to display data policy of the site
- **ISO 28258**: to display the help page on access Soils4Med data in ISO 28258 standard

2. First Access

The MEd-SIS system has a moderate user registration by default; this means that the users must be validated by the staff after the registration (Figure 3) .

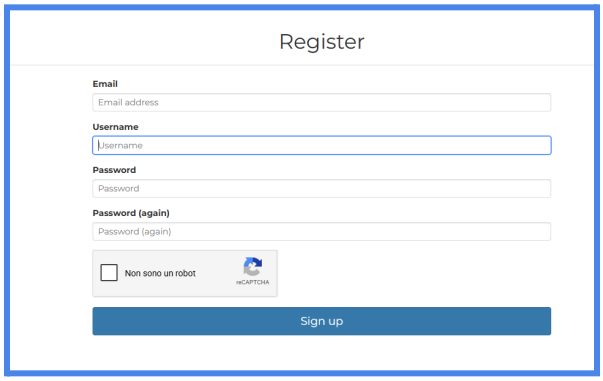


Figure 3: Registration page

Therefore, registration requires an exchange of emails between the system administrator and the user. The system administrator's contact details are available on the "Data Policy" page, accessible via the link at the bottom of the homepage.

Once registered and approved, users can access the catalogue .using the “Sign in” link in the top of the home page

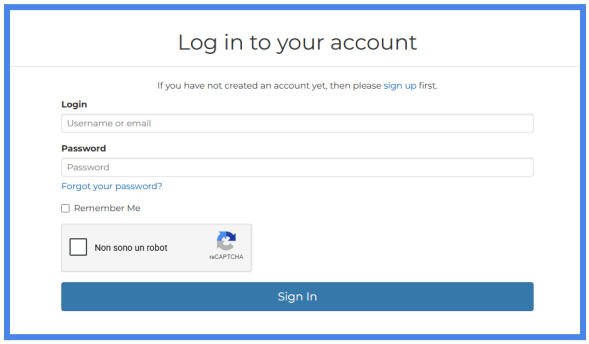


Figure 4: Sign-in page

Users can access a resource only according to the permissions granted by the data managers, who are responsible for the data. Typically, the permissions granted to a generic user include the ability to view the data and its metadata.

3. User interface

The system's main page (Figure 1) displays the project logo in the upper-left corner, which redirects to the homepage when clicked.
In the upper-right corner there are: the language selection menu (English, Deutsch, Español, Français, Italiano); the user login icon provides access to the login page. Once the user is logged in, it displays a list of available actions, as shown in Figure 5:

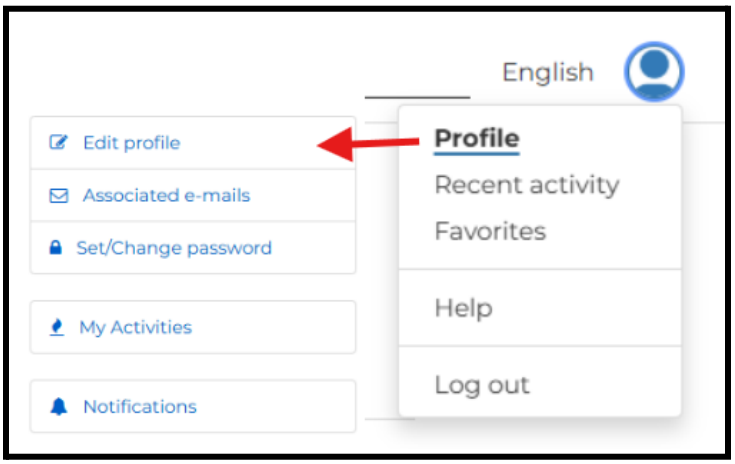


Figure 5 - User profile

- ***Profile***: displays the user's profile
- ***Recent activity***: displays the user's recent activities
- **Favorites**: displays the list of the favorites datasets
- **Help**: provides quick help on performing actions in the catalogue
- **Log out**: logs the user out of the system.

In the top left, the menus provide direct access to the resources available to the user, organized by category (Datasets, Maps, Documents, Geostories, and Dashboards) or displayed together (All Resources). These resources can also be accessed by scrolling down the page.

The resources list panel

In the center of the page header, there are quick links that provide direct access to the following resources, if made available by the data manager responsible for them:

- **Soil Maps** : Vector / Raster datasets with Pedological Classification
- **Soil Point Data** : Point vector datasets with profile and samples soil data

- **Chemical Health** : Vector / Raster datasets with soil indicators describing chemical health
- **Physical Health**: Vector / Raster datasets with soil indicators describing physical health
- **Biological Health**: Vector / Raster datasets with soil indicators describing biological health



Figure 6: List of resources and filters

The resource list can be filtered using the filter panel, accessible through the "Filter" button, as shown in Figure 6. When a filter is applied, a green dot is displayed on the "Filter" button.

From the list of available resources, clicking on the resource name opens the resource page, while clicking the three-dot menu on the right provides access to additional actions, including the preview. The preview provides an overview of the resource, including its geographic view, attributes, metadata, and other information made available for that resource (Figure 7).

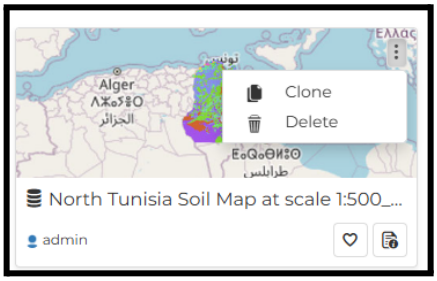


Figure 7: the panel of a resource in the list

As mentioned above, the SIS resources can be distinguished in Datasets, Maps, Documents, Geostories, and Dashboards.

All or each resource can be filtered with one or more of the following filters:

- **My resources**: resources owned by the user
- **Favorites**: resources explicitly marked as favorites by the user.
- **Featured**: resources explicitly marked by data managers or administrators as highlighted or recommended
- **Pending approval:** that has been submitted and is awaiting approval by a data manager or administrator.
- **Remote**: resources served by an external service, such as Web Map Service, ArcGIS REST service,or another GeoNode instance
- **Datasets:** set of data
    - **Vector**
    - **Raster**
    - **Time series**
    - **3D Tiles**
    - **Tabular**
- **Documents**: all available documents
- **Maps**: all available maps including those created using the Geonode Map Viewer, as well as maps created directly through MapStore-based viewer configuration files, through APIs or scripts, or imported from other Geonode instances
- **MapViewers**: only maps created using the Geonode Map Viewer
- **GeoStories**: all available Geostories
- **Dashboards**: all available Dashboards
- **Category**: filters resources based on the value assigned to the Category metadata field. Typical categories, often based on INSPIRE / ISO topic categories used in geospatial metadata standards, include things like: Boundaries, Environment, Transportation, Economy, Health, Society, Utilities / Communication
- **Keywords**: filters resources based on the keywords assigned in the metadata, typically free text tags
- **Regions**: filters resources based on metadata associated with the geographical region of the resource. This is distinct from the bounding box, which is automatically derived from the geographical extent of the data
- **Owner**: filters resources based on the user who created or owns them
- **Group**: filters resources associated with a specific GeoNode group
- **Date from**: filters resources based on their creation or upload date, showing only resources added on or after the specified date
- **Date t**o: filters resources newer than the selected date
- **Extent**: resources whose geographical area intersects with or is contained within the selected extent

Dataset Resource Panel

Dataset are geospatial resources that can be stored, managed, visualized, and shared through the platform. A dataset typically represents a spatial layer.

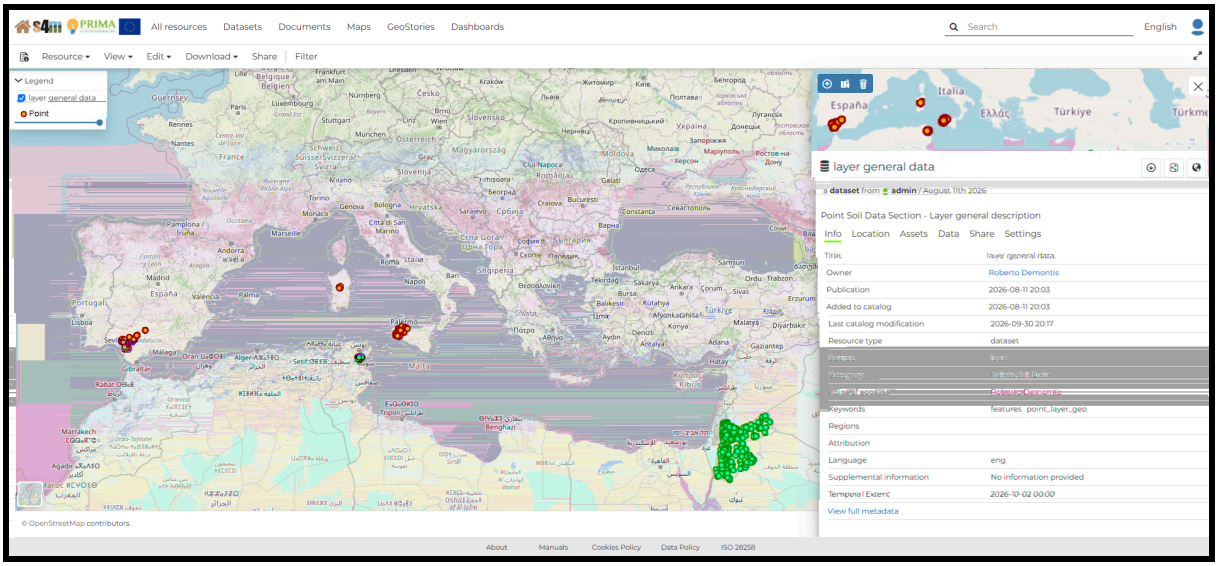


Figure 8: The dataset resource page

By clicking the **Open Property button** (top left) the user can contemplate the resource's general information, geographical location, alphanumeric attributes, and the resources linked to the dataset, either as resources that use it or as resources used by it.

If the user has "manage" permission, the dataset can be saved or deleted through the **Resources menu**.

by clicking the **View menu**, the user can access the resource's metadata and/or the alphanumeric data associated with the dataset. When clicking "view data", the advanced search icons in the upper left of the table provide the following options :

- Perform an advanced search using attribute filter, an area of interest, or a layer filter;
- filter the data displayed in the table according to the current view after zooming;
- zoom to the page extent;
- hide or show table columns;
- synchronize the map with the applied filters.

Examples of "advances search" functions are shown in Figure 9 and Figure 10



Figure 9

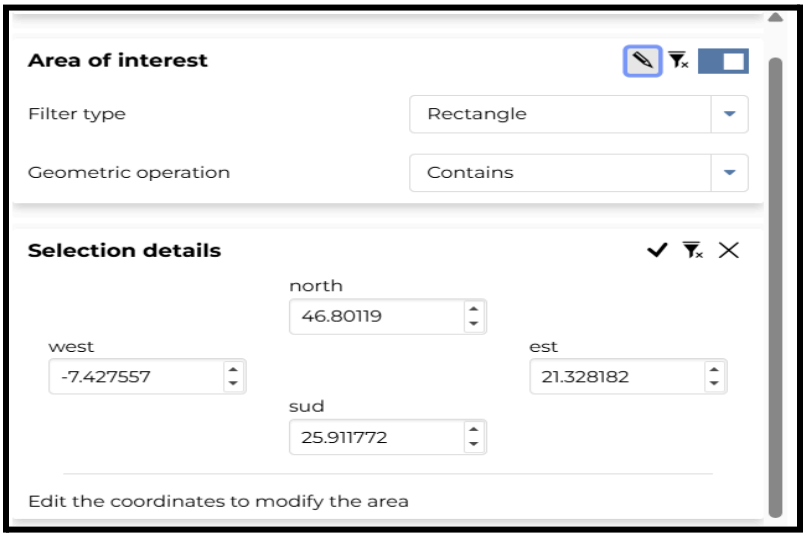


Figure 10

By clicking the **Download menu**, the user can:

- download the dataset as a compressed **.zip** file containing the corresponding shapefile, which can be opened using GIS software;
- export the dataset in different formats including GeoJSON, GML-FeatureCollection, GML-GFS 1.1.0 FeatureCollection, Shapefile, and CSV;
- export only the portion of the dataset currently displayed in the map usingCrop dataset to current viewport;
- Export only the subset defined by the applied filters using Download filtered dataset
- download the ISO metadata in .xml format (Figure 11).

The “viewport” filter types the search results to resources that fall within the current map extent i.e., the area current visible on the screen.

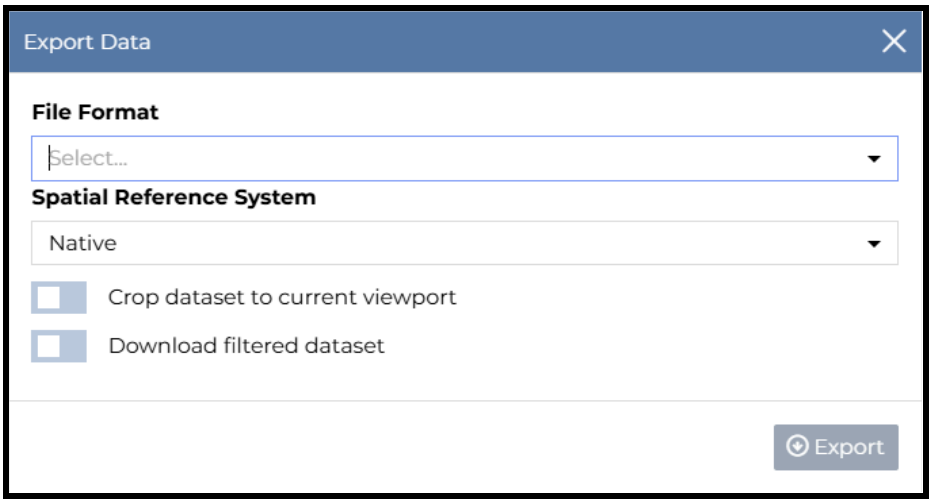


Figure 11

By clicking the **Filter menu,** the user can perform the same filtering operations available through the icons in the **View menu.** The user can perform advanced searches including attribute-based filtering, and area of interest selection. For attribute filtering, multiple conditions can be applied by clicking the "+" button to add additional conditions. (Figure 12)

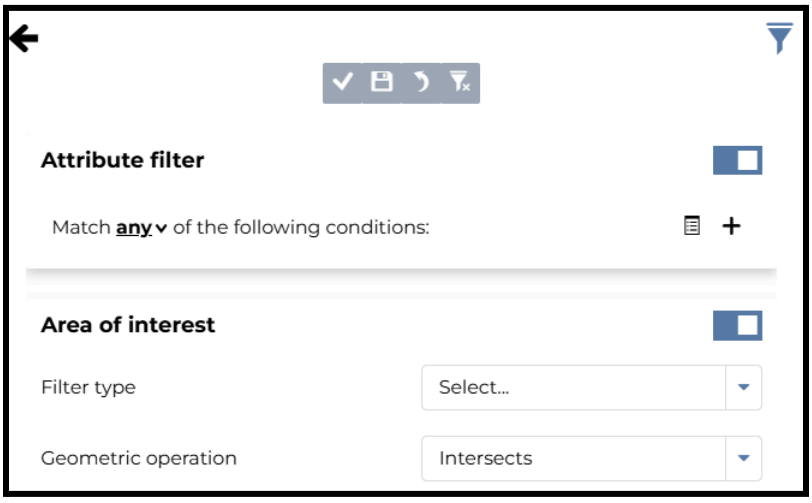


Figure 12

Map Resource Panel

Maps are compositions or visualizations built from one or more datasets (layers).
The Figure 13 shows the map resource panel

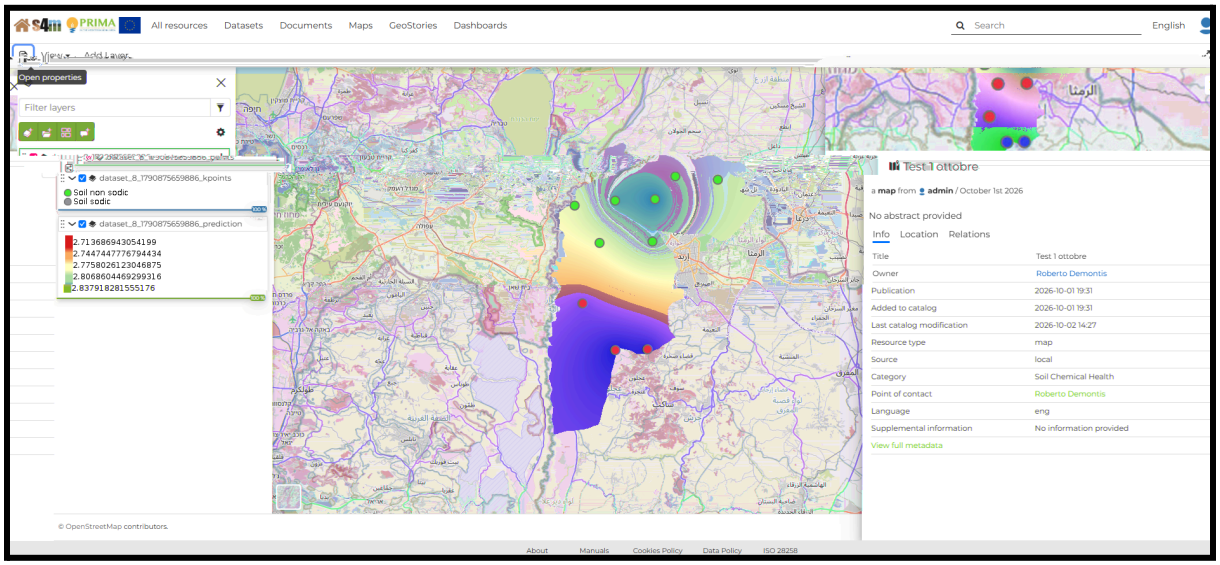


Figure 13: the map resource page

From the map resource panel users can:

- Access and view the map through the web interface
- Interact with the map by zooming, panning, toggling layers on or off
- Select a base map which is set to OpenStreetMap by default
- View the associated metadata (title, description, etc.) by clicking the **Open Property button**
- Explore the layers included in the map by turning them on/off and viewing basic layer information. In some cases, when the layers included in the map are downloadable, user can also open the attribute table, query features, and apply filters
- Open the map directly through its URL
- Share the map with others by sharing its URL. Since the map is public, the link can be accessed by others. The sharing button is located on the right-hand side of the map information panel
- Add layers by clicking the first of the four buttons on the left (Figure 13, 14 and 15). User can add a layer of the catalogue or an external layer by selecting one of the following services or formats: WMS (Web Map Service), WMTS (Web Map Tile Service), CSW (Catalog Service for the Web), TMS (Tile Map Service), XYZ Tiles, WFS

(Web Feature Service), IFC Model, and ArcGIS REST.



Figure 13

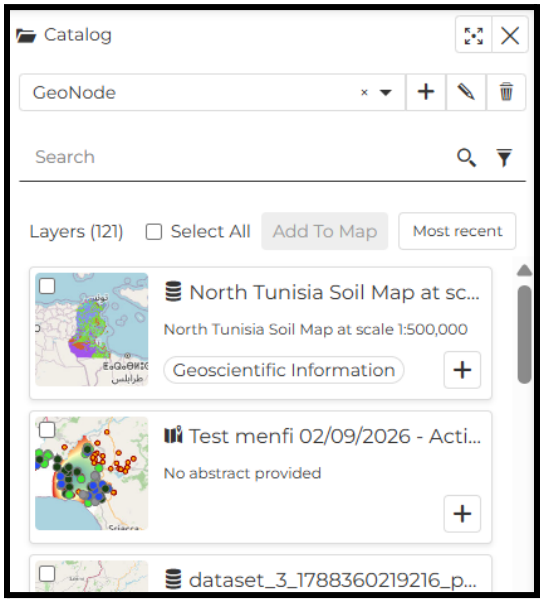


Figure 14

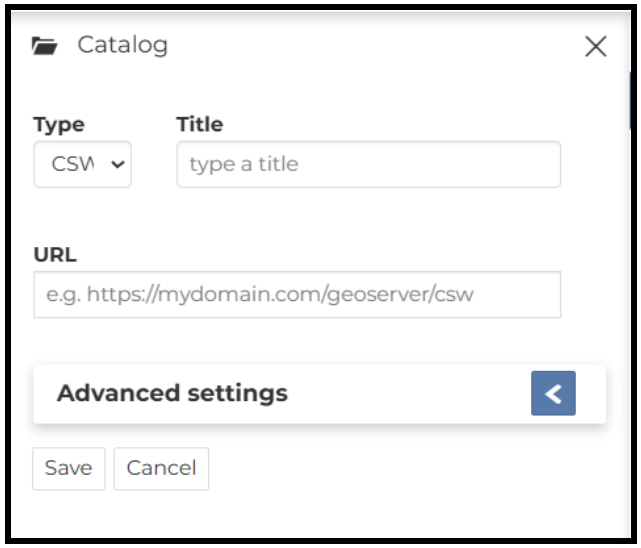


Figure 15

The Advanced settings allow users to configure how a remote layer is:

- **Loaded**: service search: automatically retrieves available layers from the selected remote service, server type: defines the type of remote server (e.g., GeoServer, ArcGIS) to ensure the appropriate requests are sent
- **Displayed**: preview: displays a preview of the layer before it is adding it), format: sets the image format (e.g., PNG, JPEG) which affects quality and transparency, single tile vs tiled: determines whether the map is requested as a single large image or as a multiple tiles. Single tile mode may provide better image quality, but can result in slower performance, tile size: sets the size of the tiles (e.g., 256×256 pixels), affecting performance and the number of requests
- **Optimized**: visibility limits: determines the zoom levels at which the layer is visible, tile grids uses the tiling scheme defined by the remote service to ensure better alignment, domain aliases: specifies alternative domains for the same service which can be useful for load balancing or managing browser connection limits
- **Interacted with**: interactive legend: allows users to interact with the legend, for example by toggling styles or classes, information sheet: defines the format of the feature information displayed when user clicks on the map, (e.g., HTML, JSON).

By clicking the second of the four buttons in Figure 13, users can add a new group of features to the current map.

By clicking the third and the fourth button shown in Figure 13, users can add widgets and annotations to the current map. If the user has edit permissions, annotations become part of the map. For a generic user without edit permissions, annotations are usually temporary and are lost when the page is refreshed.

Document resource panel

Documents are used to upload, store and share non-spatial files through the platform, such as PDFs (reports, studies), Word documents, Excel spreadsheets, Images, Presentations.

Users may be able to open, download, edit, or manage documents. For example, a PDF can be previewed and downloaded (depending on permissions). As with other SIS resources, documents may have different access and management options depending on the user's permissions. Generic users can usually:

- **View** the document and its metadata
- **Download** the document and its ISO metadata

The SIS includes documents related to Soil Point Data, Soil Maps, Soil Indicators and the User Manuals.

Geostory resource panel

GeoStories are a special content type used to create interactive narratives that combine maps and multimedia content. They are essentially web-based storytelling pages that can integrate: maps already available in the SIS, text such as descriptions, and explanations, images and other media, embedded content.

If the GeoStory is public, a registered, non-privileged user can:

- **View** the story and its metadata while following the narrative flow,
- **Interact** with embedded maps to explore geographic content
- **Share** the GeoStory via its URL using the sharing button located on the right hand side

Dashboard resource panel

Dashboards are interactive pages used to visualize and analyze geospatial data through charts, widgets, and maps. They are data visualization tools that combine different elements, including: Maps, Charts (bar, pie, line, etc.), Indicators /counters, Filters and widgets. These elements allow users to summarize and explore data in a dynamic and visual way.

If the user has the necessary permissions to access the resources used in the dashboard, they can:

- **View** the dashboard and its metadata including charts, maps, and widgets
- **Interact** with the visualizations, for example by exploring chart bars, maps
- **Apply filters** such as time-based or attribute-based filters, and see the visualizations update dynamically.

4. How to request access to data

To access the soil data, users must be registered in the system and submit a request to the data managers—specifically, the SIS staff—using the contact details provided on the "Data Policy" page shown in Figure 16.



Figure 16

The request must include

- the **resources' names** for:
    - A Soil indicator dataset resource name (see the document "MED-SIS-Manual-Soil_Indicators.pdf" in the Help page) or a set of map names;
    - A Point Soil data section dataset resource name (see the document "MED-SIS-Manual-Points_Soil_Data.pdf" in the Help page) or a set of names of map names
    - A set of Soil indicators and Point Soil data section datasets to be used to create a map
- The **area of Interest**, which can also be specified using the name of an administrative division

The request may also include the following parameters:

- **observation period:** start and end dates or all available dates
- **depth:** upper and lower depth limits or all available depths
- **type**: Auger hole, Minipit or Profile or all types
- field survey **method**: the method used to carry out the field survey or all methods:

Identifier	Code	Description
SURVEY_METHODS:CP	CP	method: CPCS;
SURVEY_METHODS:FA	FA	method: FAO
SURVEY_METHODS:FA06	FA06	method: FAO; year: 2006
SURVEY_METHODS:FA52	FA52	method: FAO; year: 1952
SURVEY_METHODS:FA77	FA77	method: FAO; year: 1977
SURVEY_METHODS:FA90	FA90	method: FAO; year: 1990
SURVEY_METHODS:US	US	method: USDA soil survey manual
SURVEY_METHODS:US18	US18	method: USDA soil survey manual ; year: 2018
SURVEY_METHODS:US52	US52	method: USDA soil survey manual ; year: 1952
SURVEY_METHODS:US62	US62	method: USDA soil survey manual ; year: 1962
SURVEY_METHODS:US75	US75	method: USDA soil survey manual ; year: 1975
SURVEY_METHODS:US93	US93	method: USDA soil survey manual ; year: 1993
SURVEY_METHODS:WR	WR	method: WRB;
SURVEY_METHODS:WR22	WR22	method: WRB; year: 2022

Figure 17

- **Project:** the Identifier or name of the project, or survey activity/campaign, within which the point soil data were generated or all projects
- laboratory **method:** the method used for the laboratory measurements, no specific method or all available methods,depending on the measurements involved.

If the requested dataset concerns a soil indicator, the user can also request an interpolation of the measurements over the selected area of interest. The interpolation will be performed using ordinary kriging.

After the requested datasets have been created and **if the data managers evaluation is ok**, the datasets:

- filtered point data,
- Interpolated data
- aggregated point data used for the interpolation

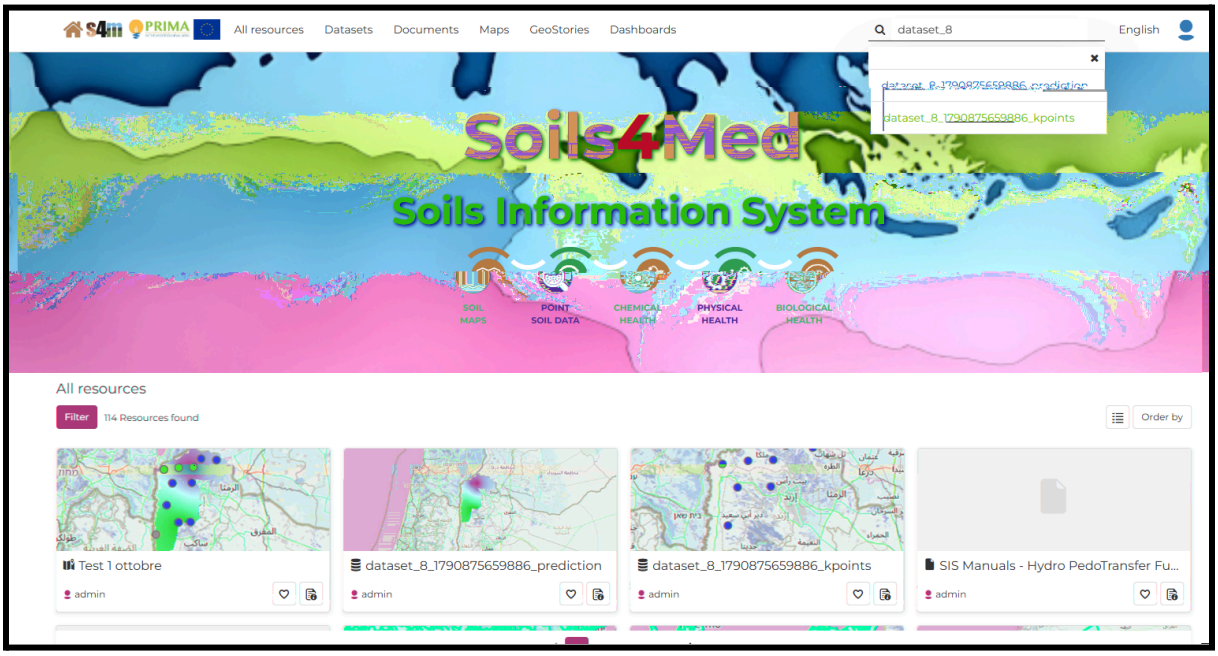


Figure 18

3. API

The MED-SIS uses API managing Catalogue resources.

A dedicated API is available for managing soil data, accessible from the back office.

For security reasons, access to the APIs is restricted to the MED SIS Staff!

The data generated by the Soils4Med project will be open and accessible via specialized APIs. These APIs, which comply with the ISO 28258 standard, are described in the document MED-SIS-MANUAL-ISO_28258.pdf.